Arko Das

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EDUCATION

University College London | London **Sep 2023 - May 2027**

MEng Chemical Engineering (First-Class Honours, 82% Average)

- Capstone Plant Design: Used ASPEN Plus to design a urea synthesis flowsheet (46,000 t/y), applying pinch analysis and heat integration, LCA, and techno-economic analysis to optimise energy efficiency and commercial viability.
- Engineered a high-pressure falling-film CO_2 stripper for urea synthesis, validating and optimising reactive equilibria; applied HAZOP, P&ID and control systems review and LCA to ensure process safety and commercial viability.
- Minor: Programming for Data Science - built interactive data dashboards in Python using Dash and SQLite, applying object-oriented design and Pandas for analysis.

Dame Alice Owens School | Potters Bar, UK **Sep 2016 - Jun 2023**

- A Levels: Mathematics (A*); Chemistry (A*); Physics (A*). Mentor, Prefect, and weekly Science Club lead.
- GCSEs: 7 Grade 9-8s, including Mathematics (9) and English (8), and 3 Grade 7s. Completed Bronze DofE award.

EXPERIENCE

Hippos Exoskeleton | *R&D and Commercial Consultant* | London **Sep 2025 - Jan 2026**

- Delivered 11 technical proposals directly to the founders of a VC-backed sports injury prevention startup, evaluating sensor materials, process costs, and cycle stability across inflation and sensing workstreams.
- Ran iterative lab trials on soft polyurethane composites, controlling mixing rates and dispersion kinetics to replace wire-based sensors.
- Conducted cost-benefit analysis across material and process variants, guiding R&D investment decisions ahead of a Series A round.

UCL Racing | *Shell Hypermile Hydrogen Engineer* | London **Sep 2024 - Present**

- Analysed transient energy and GPX data in MATLAB and Python to develop predictive race strategy models.
- Characterised H_2 utilisation on a Horizon H-200 PEM stack; sensitivity analysis improved system efficiency by 23%.
- First UK team to complete the Shell Hypermile hydrogen fuel cell challenge, scoring 364 km/m³ and placing 1st in Carbon Footprint and 2nd in Safety.

Private Mathematics Tutor | London **Sep 2022 - Jun 2025**

- Tailored lessons for GCSE and A-Levels, achieving grade improvements to 9 and 8 in exams across all 6 students.

BP | *Energy Insight Programme* | London **Sep 2023**

- Analysed CAPEX and logistics constraints across 6 case studies to quantify the adoption of alternative energies.
- Used economic models to maximise ROI for solar array designs in various environmental and social conditions.

LEADERSHIP & INITIATIVES

Aramco Ecothon | *Hackathon* | London **Apr 2026**

- Designed a modular electrochemical process converting SO_2 emissions into green hydrogen and fertiliser within a 3-day competition, backed by a techno-economic model projecting an £18M NPV and 17% ROI.
- Secured 1st place among teams from all years of Chemical Engineering, defending the proposal to Aramco judges.
- Previously placed 2nd (2024), pitching novel DNA data storage solutions for big data challenges in Oil and Gas.

UCL Ramsay Chemical Engineering Society | *Social Secretary* | London **Mar 2024 - Present**

- Managed a £5,000 budget across 16 events including professor-led career talks, doubling year-on-year turnout.
- Led industry outreach, negotiating in-depth visits with companies such as ExxonMobil and Itero for 100+ students.
- Deployed and managed a cloud-hosted server and webapp infrastructure for 60+ society members using Linux, SSH, and Oracle Cloud, reducing reliance on costly third-party hosting.

UCL Department of Chemical Engineering | *Department Liaison* | London **Sep 2025 - Present**

- Collaborated with faculty to consolidate induction resources across Moodle, the departmental intranet, and the student handbook, reaching 200+ students.
- Independently developed a family mentorship and careers structure, matching senior and junior students.

TECHNICAL SKILLS & LANGUAGES

- Proficient in ASPEN Plus, gPROMS, Python, MATLAB, GAMS, and Linux/SSH; Excel and Power BI.
- Independently completed courses in nanotechnology (Duke University), data analytics, and Python programming (Johns Hopkins University; University of Helsinki).
- Native in English; Fluent in Bangla; Working Proficiency in Spanish.